Is Poisson Good Enough?

Evaluating Queueing Models for Urban Mobility Systems

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Abstract

We test how two Boston mobility systems depart from Poisson arrival assumptions and quantify the resulting error in queueing predictions. Using four months of data from Bluebikes at Kendall/MIT and MIT Vassar St (user-driven, finite-capacity) and the MBTA Red Line at Kendall/MIT (schedule-driven), we reject exponential inter-arrival times for every system ($p \approx 0$ under Kolmogorov–Smirnov, Anderson–Darling, and χ^2) but in opposite directions: Bluebikes is overdispersed (CV = 1.75–1.90, best fit Weibull) and MBTA is underdispersed (CV = 0.63–0.71, best fit log-normal). Analytical Poisson queueing underestimates Bluebikes wait by roughly an order of magnitude and Erlang B blocking at Kendall/MIT by 55-fold, while overestimating MBTA wait by 5–12×. The 55× blocking gap persists even when the simulation uses empirical arrivals, isolating service-process non-stationarity as a separate error source that arrival-process correction alone cannot address.

1 Introduction

Queueing theory in transportation is built on the Poisson arrival assumption. Closed-form results for the M/M/c and M/M/c/c models [1, 2] give analysts tractable estimates of wait time and blocking probability. These models inform some operational decisions directly — bike-share dock counts and rebalancing cadence are a clear case — and provide back-of-envelope wait estimates for studies of transit user experience and transfer planning, while transit infrastructure sizing is governed by demand forecasts and operating constraints rather than M/M/1 wait. Real urban mobility arrivals are shaped by timetables, time-of-day demand, and disruptions, so the assumption is approximate. The practical question is not whether arrivals are Poisson — they almost never are — but how much queueing predictions suffer when the assumption fails, and whether the residual error matters for the decision the prediction feeds.

We study two Boston systems whose arrival processes are structurally opposite. Bluebikes is a user-driven bike-share with finite-capacity docks; the MBTA Red Line at Kendall/MIT runs a published timetable on an effectively uncapacitated platform. We ask: (1) how far do the observed arrivals deviate from Poisson; (2) how accurately do Poisson-based models predict wait and blocking; and (3) under what conditions is Poisson a defensible engineering approximation?

Our contributions are: quantitative Poisson-deviation measurements for both systems; a side-by-side comparison of analytical Poisson queueing against discrete-event simulation driven by the empirical and best-fit arrival distributions; identification of a 55× Erlang B blocking gap at Kendall/MIT that arrival-process correction alone cannot close, isolating service-process non-stationarity as a distinct error source; and a decision-threshold framework that evaluates Poisson's error by *direction*, *magnitude*, and whether the prediction is on the decision path at all — the

user-side blocking under-prediction is binding, while the MBTA wait over-prediction is large in relative terms but does not feed any operational decision.

2 Data

Bluebikes. Public trip data, September–December 2025, at M32004 Kendall/MIT (23 docks, transit hub) and M32042 MIT Vassar St (53 docks, campus dormitory area). An arrival is a trip-end timestamp. After operating-hours restriction, disruption-day exclusion, and fullness exclusion (below), $N = 9{,}566$ (Kendall/MIT) and 19,300 (MIT Vassar).

MBTA Red Line. LAMP performance data [3] for Kendall/MIT (`place-knncl`) in the same window, Northbound ($N = 20{,}366$) and Southbound ($N = 20{,}880$). An arrival is a train-at-platform event. Overnight gaps are excluded.

Fullness exclusion. Observed Bluebikes arrivals are censored when a station is at capacity. We reconstruct station inventory by forward event replay with bookkeeping clamps for unlogged operator rebalancing, and flag a full-capacity interval only when a natural $C_s - 1 \to C_s$ trip-end transition is followed by a trip-start with no intervening trip-end; IATs overlapping any committed interval are excluded from λ, μ, and distribution fitting. Kendall/MIT is full 3.73% of operating time and MIT Vassar St 2.88%; we use these as the validation target for Erlang B blocking (§4.3). Algorithm in Appendix A.1.

Reproducibility. Code, derived data, and figures at `https://github.com/taka1005/transportation-project` (MIT license).

3 Methods

Arrival-process characterization. For each system we compute the mean, standard deviation, coefficient of variation ($\mathrm{CV} = \sigma/\mu$), and skewness of the inter-arrival time, and the index of dispersion ($\mathrm{IoD} = \mathrm{Var}[N]/\mathbb{E}[N]$) of arrival counts in 15-, 30-, and 60-minute windows. $\mathrm{CV} = \mathrm{IoD} = 1$ are the Poisson values. We fit exponential, log-normal, Weibull, and gamma by maximum likelihood and rank them by AIC. Fits are tested with Kolmogorov–Smirnov (KS), Anderson–Darling (AD), and Pearson χ^2 statistics.

Queueing models. Bluebikes: M/M/c (infinite-queue idealisation; real stations reject rather than queue) with c the dock count,

$$W_q = \frac{C(c,a)}{c\mu - \lambda}, \quad a = \lambda/\mu,\ \rho = a/c, \tag{1}$$

where $C(c,a)$ is the Erlang C probability of queueing, and M/M/c/c giving the Erlang B blocking $B(c,a) = a\,B(c-1,a)/(c + a\,B(c-1,a))$ with $B(0,a) = 1$. MBTA: M/M/1 with service = dwell time. Bluebikes μ is estimated via Little's Law [4] from the average non-full dock occupancy. Derivations in Appendix D.

Discrete-event simulation. We implement each system in SimPy [5] with three arrival generators: Poisson ($\mathrm{Exp}(1/\lambda)$), empirical i.i.d. bootstrap from the observed IATs, and best-fit sampling from the MLE Weibull (Bluebikes) or log-normal (MBTA). Each configuration uses 20,000 arrivals with a

2,000-arrival warmup and 10 replications (95% CI via Student's t).

4 Results

4.1 Arrivals reject Poisson in opposite directions

Every system rejects exponential inter-arrivals at $p \approx 0$ under KS, AD, and χ^2 (Appendix B.1). Table 1 summarises the direction: Bluebikes is overdispersed (CV 1.75–1.90, IoD 2–8); MBTA is underdispersed (CV 0.63–0.71, IoD 0.4–0.6). The best parametric fit is Weibull for Bluebikes (shape $c \approx 0.72$, monotone-decreasing hazard) and log-normal for MBTA (shape $s \approx 0.5$), with AIC improvements of 2,167 to 17,833 over exponential. MBTA in fact departs from exponential *farther* in KS distance (0.27–0.29) than Bluebikes (0.13–0.14): regularity is as extreme a Poisson violation as burstiness.

Table 1: Inter-arrival-time summary statistics. CV < 1: underdispersion; CV > 1: overdispersion.

System	N	Mean IAT (s)	CV	Skew	IoD (30 min)
BB Kendall/MIT	9,566	661	1.75	6.9	3.2
BB MIT Vassar	19,300	380	1.90	7.7	4.7
MBTA Northbound	20,366	390	0.71	3.3	0.58
MBTA Southbound	20,880	378	0.63	4.5	0.52

Within-day segmentation (Appendix B.2, Table 4; empirical CDFs with fits in Appendix C, Fig. 7) attributes the Bluebikes CV to non-stationarity: Kendall/MIT falls to CV ≈ 1.0 at midnight and peaks near 2.0; MBTA stays below 1 at every hour. A single-window Poisson hypothesis is thus rejected *more* strongly at peak hours, not less.

4.2 Queueing predictions diverge in magnitude and direction

Table 2 compares analytical Poisson predictions to DES under the three arrival generators; Poisson-DES cross-checks the analytical value and empirical-DES is the benchmark. Bluebikes W_q is an infinite-queue idealisation (caption); the observable Bluebikes metric is blocking (§4.3).

Table 2: Average queueing delay W_q (seconds). Bluebikes values are under an infinite-queue idealisation (real stations reject rather than queue), reported for common-abstraction comparison with MBTA. 95% CIs on DES values within ±10%.

System	Analytical	Poisson DES	Empirical DES	Best-fit DES
BB Kendall/MIT	0.8	1.4	9.4	10.6
BB MIT Vassar	≈ 0	≈ 0	≈ 0	≈ 0
MBTA Northbound	20.1	20.5	4.2	4.3
MBTA Southbound	10.7	10.9	0.9	1.0

At Kendall/MIT Poisson M/M/c predicts $W_q = 0.8$ s against empirical 9.4 s — about an order-of-magnitude underestimate. For MBTA the sign flips: 20.1 s vs 4.2 s Northbound (5× over), 10.7 s vs 0.9 s Southbound (12× over). The Northbound–Southbound spread is driven by dwell-time asymmetry at Kendall/MIT (mean dwell 82 s vs 61 s, reflecting heavier morning-peak boardings on the inbound platform), not by arrival-process differences. Best-fit-DES tracks empirical-DES

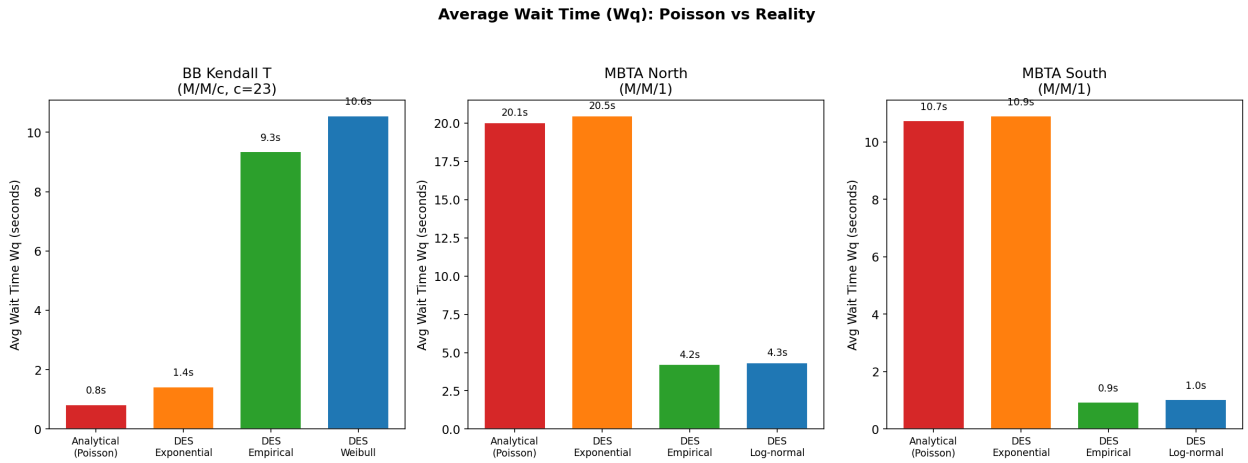

Figure 1: W_q across systems (log y-axis). Bluebikes Kendall/MIT is the only case where Poisson underpredicts; both MBTA directions overpredict. Best-fit-DES tracks empirical-DES to within a second.

to within a second everywhere, so a Weibull or log-normal fit retains enough of the arrival-process information to reproduce the queueing outcome. MIT Vassar St is a W_q null case under every model (53 docks drive the queueing wait to zero), yet §4.3 shows it is *not* a blocking null case once the observed fullness rate is reconstructed properly.

4.3 The Erlang B blocking gap isolates service non-stationarity

At Kendall/MIT the Erlang B prediction for blocking is 0.068% against an observed full-capacity rate of 3.73%: a 55× gap. Finite-capacity DES with *empirical* IATs gives 0.54%, closing only about a seventh of the gap. At MIT Vassar St the gap is qualitatively similar (Erlang B $\approx 0\%$ vs observed 2.88%; empirical-DES produces no blocking events). Arrival correction alone therefore cannot explain the observed blocking at either station; the residual is attributable to service-process non-stationarity (dock occupancy durations swell during rush-hour departures and collapse midday). A time-varying μ interacting with a time-varying λ cannot be captured by an M/M/c/c model or by empirical-arrivals-with-exponential-service DES. The MBTA has no analogue at $\rho = 0.15$–0.20.

5 Discussion

Poisson's error has three orthogonal axes: *direction* (over- vs. under-estimation), *magnitude relative to the decision threshold*, and *whether the prediction is on the decision path at all*. The two systems we study fall into qualitatively different regimes.

User-side, on the decision path: Bluebikes blocking. Poisson under-estimates Kendall/MIT blocking by 55-fold. Bike-share planners do use queueing-style reasoning to size docks and tune rebalancing cadence, so a Poisson target would miss service failures by an order of magnitude; users find full stations and abandon trips. Bursty arrivals (CV 1.75–1.90) average away the tail events that drive blocking. This is the binding error in our study.

Off the decision path: MBTA wait. MBTA at $\rho = 0.15$–0.20: Poisson overestimates W_q by 5–12×. The relative error is large, but transit infrastructure is not sized from per-train W_q: platform

length is inherited capital, service frequency is set from demand and operations constraints, and crowding (not queueing wait) is the customer metric. 20 s vs 4 s are both below the minute-scale resolution of any transit decision.

Error direction is predictable from structure. User-driven, unscheduled systems produce overdispersed arrivals and are under-predicted by Poisson; timetabled systems produce underdispersed arrivals and are over-predicted. Evaluate direction first, then magnitude against the specific decision.

6 Limitations

Arrival censoring at full Bluebikes docks. Observed arrivals at a full station are zero by construction, so our λ estimate is a lower bound. Mellou and Jaillet [6] give a canonical recovery method; we do not apply it because substituting estimated arrivals would confound a real Poisson deviation with an estimator residual.

DES destroys inter-arrival autocorrelation. Empirical-DES samples IATs i.i.d., preserving the marginal distribution but not the sequential structure imposed by the timetable. This is why MBTA Southbound empirical-DES reports $W_q = 0.9$ s rather than essentially zero.

Scope. Two Bluebikes stations, one MBTA station, four months. The structural argument should generalise; specific magnitudes should not.

Service-process non-stationarity unmodelled. The residual 55× blocking gap is attributable to time-varying service that we diagnose but do not fit.

Inventory reconstruction uncertainty. I_s is reconstructed forward from trip events and carries a bounded rebalancing bias between clamps; our retract rule (Appendix A.1) biases toward omitting short full spans, so observed fullness and the 55× gap are conservative lower bounds.

7 Future Work

Re-running §4.1 and §4.3 on Bluebikes arrivals reconstructed with the Mellou–Jaillet estimator [6] would separate censoring-induced from real non-Poissonness and quantify how much of the 55× blocking gap is attributable to arrivals vs. service. An autocorrelation-preserving arrival generator (time-series bootstrap or Markov-modulated Poisson) should drive MBTA empirical-DES W_q toward zero. Fitting an hour-of-day-indexed service distribution for Bluebikes and recomputing Erlang B with time-varying μ would directly address the service non-stationarity finding.

8 Conclusion

Poisson is good enough only when its error is small relative to the decision it feeds, and only when the prediction is on a decision path at all. Bluebikes under-estimates blocking by 55× and feeds dock-count and rebalancing choices, so the error is binding; MBTA over-estimates wait by 5–12× but per-train W_q is not how transit is planned, so the same magnitude is harmless. The Bluebikes gap persists under empirical arrivals, isolating service-process non-stationarity as an error arrival correction cannot repair. Evaluate Poisson by direction, magnitude, and decision path.

References

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[3] Massachusetts Bay Transportation Authority, "LAMP: Light, heavy, and commuter rail performance data," https://performancedata.mbta.com/, 2025, accessed 2026-04.

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[5] Team SimPy, "SimPy: Discrete event simulation for Python," https://simpy.readthedocs.io/, 2020.

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A Methodology details

A.1 Bluebikes fullness-exclusion algorithm

Inventory reconstruction. Let $I_s(t) \in \{0, 1, \ldots, C_s\}$ denote the reconstructed inventory at station s at time t, where C_s is the station capacity ($C_{\text{Kendall}} = 23$, $C_{\text{Vassar}} = 53$). We process the chronological event stream of trip-ends at s ($+1$) and trip-starts from s (-1) in a single forward pass, starting from $I_s = \lfloor C_s/2 \rfloor$ as an initial guess (real-time station status such as GBFS was not available for the study window). Operator rebalancing (truck pickups and drop-offs) is not recorded in the public dataset, so we infer it by clamping: whenever the running count would exceed C_s, we attribute the excess to a prior rebalancing pickup and set $I_s = C_s$; whenever it would fall below zero, we attribute it to a prior drop-off and set $I_s = 0$. A clamp is a bookkeeping correction for unlogged rebalancing; it is not itself evidence of a full/empty state.

Full-capacity windows. We treat "full" as a *state* rather than a per-event property. A candidate window opens at a trip-end that brings I_s from $C_s - 1$ to C_s via natural increment (not clamp). The window is committed to the exclusion set only if the next event at s is a trip-start (departure), which then closes the window. If the next event is instead another trip-end, the successful trip-end is physical evidence that a slot was available at that instant, so the candidate window is *retracted entirely*: we lack evidence that the station was continuously full across the interval, and we do not retain any partial subinterval. This rule is the reason the observed fullness rate in §2 should be read as a conservative lower bound; see also §6.

Exclusion rule. An inter-arrival time (a_i, a_{i+1}) is excluded from λ, μ, and distribution fitting if the interval $[a_i, a_{i+1}]$ overlaps any committed full window; the non-censored arrivals a_i and a_{i+1} are themselves retained for use in adjacent IATs. Committed windows do not overlap by construction (a window ends at a trip-start that lowers I_s to $C_s - 1$, so any subsequent natural entry requires a later trip-end). For Kendall/MIT, 668 IATs (6.5% of the uncorrected sample) are excluded, giving $N = 9{,}566$. For MIT Vassar St, 148 IATs (0.8%) are excluded, giving $N = 19{,}300$.

Robustness. The forward pass is anchored at every clamp: after the first clamp event, I_s is pinned to a boundary regardless of the initial guess, so the effect of $\lfloor C_s/2 \rfloor$ decays within the first clamp (typically within the first days of operation). Because the full flag is defined by a natural $C_s - 1 \to C_s$ transition under the retract-on-intervening-trip-end rule, clamp-driven drift corrections do not inflate the fullness rate. The residual uncertainty in I_s between two clamp events (a constant additive bias bounded by the untracked rebalancing volume) is discussed as a limitation in §6.

A.2 Service-rate estimation via Little's Law

For Bluebikes we treat each dock slot as a server and define the service time as the elapsed time between a bike arriving at the dock and the next departure of that slot. We estimate μ from reconstructed inventory using Little's Law:

$$\mu = \frac{\lambda \, \bar{W}_{\text{slot}}}{L_{\text{non-full}}} \cdot \frac{1}{c}, \tag{2}$$

where $L_{\text{non-full}}$ is the mean number of occupied docks during non-full periods, c is the dock count, and $\bar{W}_{\text{slot}}$ is the resulting per-slot average service time. After fullness correction the estimated mean service times are 7,367 s at Kendall/MIT and 6,670 s at MIT Vassar St, yielding per-server utilisations of $\rho = 0.48$ and $\rho = 0.33$ respectively. For MBTA the service time is the observed platform dwell time; ρ ranges from 0.15 to 0.20.

A.3 Discrete-event simulation

The SimPy implementation models each dock (Bluebikes) or platform (MBTA) as a `simpy.Resource` with capacity equal to the physical server count. An arrival either seizes an available server immediately or, in the infinite-queue variant, joins a FIFO queue; in the finite-capacity variant (M/M/c/c, Bluebikes only) an arrival finding all servers busy is immediately rejected and counted toward the blocking probability. Service times are sampled from the exponential distribution with rate μ estimated per the previous subsection. Each run consists of 20,000 arrivals; the first 2,000 are treated as warm-up and excluded from statistics. A warm-up length of 2,000 was chosen empirically from the convergence of the running mean of W_q: the cumulative mean stabilises well before that cutoff in every system. Each configuration is replicated 10 times with independent seeds, and 95% confidence intervals are computed from the 10 replication means via the Student's t distribution.

A.4 Arrival generators

The three arrival generators share a single sampling interface and differ only in the distribution they draw from:

- **Poisson**: $T_i \sim \text{Exp}(\lambda)$ with λ set from the observed post-exclusion arrival rate.
- **Empirical**: T_i sampled i.i.d. with replacement from the list of observed inter-arrival times. This preserves the marginal IAT distribution exactly but destroys any sequential (autocorrelation) structure.
- **Best-fit**: T_i sampled from Weibull(c, scale) for Bluebikes or Log-Normal(s, scale) for MBTA, with parameters taken from the MLE fits in Appendix B.1.

B Extended results tables

B.1 Goodness-of-fit statistics

Table 3 reports the Kolmogorov–Smirnov, Anderson–Darling (exponential only, standard form), and Pearson χ^2 statistics for each distribution, together with maximum-likelihood parameter estimates and AIC. In every row every test rejects the null at $p < 10^{-10}$; the rankings are therefore relative.

B.2 CV by hour-of-day

Table 4 tracks CV through the operating day. Bluebikes Kendall/MIT approaches CV = 1.0 between 23:00 and 01:00 and peaks at CV = 2.4 at 18:00; MBTA stays within 0.48–0.84 throughout service. MIT Vassar St is consistently overdispersed (1.03–2.16) with less hour-to-hour swing than Kendall/MIT, reflecting a more uniform demand profile.

Table 3: Goodness-of-fit statistics by system and candidate distribution. All tests reject each null at $p \approx 0$. Best AIC per system in bold; χ^2 uses 30 equal-probability bins.

System	Distribution	KS stat	AD stat	χ^2	AIC
BB Kendall/MIT	Exponential (scale=674)	0.131	479.5	1873.5	136,583
	Log-normal (s=1.66, sc=246)	0.067	—	862.0	135,088
	Weibull (c=0.72, sc=533)	0.021	—	87.5	**134,416**
	Gamma (a=0.61, sc=1101)	0.051	—	237.9	134,804
BB MIT Vassar	Exponential (scale=380)	0.138	1081.6	3889.2	267,503
	Log-normal (s=1.61, sc=139)	0.061	—	1693.7	263,220
	Weibull (c=0.71, sc=297)	0.027	—	299.9	**262,613**
	Gamma (a=0.61, sc=620)	0.057	—	1254.8	263,717
MBTA Northbound	Exponential (scale=388)	0.267	1888.6	11,364.2	283,526
	Log-normal (s=0.57, sc=325)	0.046	—	2068.7	**270,649**
	Weibull (c=1.60, sc=437)	0.117	—	5662.4	276,492
	Gamma (a=2.97, sc=130)	0.066	—	3950.0	273,433
MBTA Southbound	Exponential (scale=381)	0.294	2734.0	15,628.9	289,901
	Log-normal (s=0.49, sc=334)	0.028	—	1027.5	**272,068**
	Weibull (c=1.76, sc=430)	0.115	—	5061.3	279,469
	Gamma (a=3.97, sc=96)	0.054	—	2181.5	274,864

Table 4: Coefficient of variation by hour-of-day. Hours with $N < 20$ omitted.

Hour	05	06	07	08	09	10	11	12	13	14	15	16	17	18
BB Kendall/MIT	—	—	1.30	2.00	1.47	1.58	1.48	1.44	1.44	1.52	1.35	1.26	1.67	2.38
BB MIT Vassar	—	1.40	1.03	1.41	1.41	1.52	2.16	1.37	1.71	1.59	1.50	1.68	1.96	1.55
MBTA North	0.48	0.66	0.68	0.76	0.74	0.84	0.70	0.72	0.66	0.72	0.69	0.76	0.74	0.70
MBTA South	0.35	0.56	0.67	0.72	0.71	0.84	0.59	0.62	0.57	0.57	0.57	0.69	0.61	0.62

Hour	19	20	21	22	23	00
BB Kendall/MIT	1.29	1.16	1.18	1.15	1.08	1.01
BB MIT Vassar	1.54	1.43	1.76	1.56	1.25	1.40
MBTA North	0.71	0.69	0.68	0.63	0.62	0.67
MBTA South	0.64	0.61	0.70	0.53	0.56	0.59

B.3 CV by day-of-week and IoD segmentation

Table 5 shows the CV weekday-vs-weekend pattern. MBTA is clearly more regular on weekends (CV drops from ≈ 0.74 to ≈ 0.55) — the schedule is thinner but more consistent. Bluebikes Kendall/MIT is more regular on weekends (lower commute burstiness); MIT Vassar St is slightly *more* overdispersed on weekends (leisure trips scatter across the day and produce long gaps).

B.4 Queueing outcomes with confidence intervals

Table 6 extends the main-text Table 2 with 95% confidence intervals over 10 replications and the blocking probabilities for Bluebikes. Confidence intervals on W_q are narrow ($\pm$3%–10% of the point estimate) except at MIT Vassar St, where all values are effectively zero, and at Kendall/MIT where the Poisson-DES CI is wider because the $\rho_{\text{server}} \approx 0.48$ regime produces only occasional queue events per replication. The Erlang B prediction at Kendall/MIT (0.068%) is dwarfed by both the empirical-DES finite-capacity result (0.54%) and the observed full-capacity rate (3.73%). MIT

Table 5: Coefficient of variation by day-of-week, and peak/off-peak IoD (60-min window).

System	Mon	Tue	Wed	Thu	Fri	Sat	Sun	CV Wkday	CV Wknd	IoD Peak	IoD Off-Pk
BB Kendall/MIT	2.13	1.51	1.73	2.08	1.53	1.61	1.49	1.79	1.56	4.4	2.6
BB MIT Vassar	1.87	1.95	1.66	1.79	1.86	1.81	2.24	1.83	2.01	8.4	5.1
MBTA North	0.76	0.78	0.75	0.70	0.70	0.61	0.63	0.74	0.62	0.48	0.55
MBTA South	0.70	0.70	0.63	0.64	0.59	0.53	0.57	0.65	0.55	0.50	0.54

Vassar St mirrors the gap qualitatively (Erlang B and empirical-DES both ≈ 0, observed 2.88%).

Table 6: W_q (seconds, mean $\pm$ half-width of 95% CI) and blocking probability (Bluebikes only). Bluebikes W_q is an infinite-queue idealisation; the real system rejects rather than queues, so the physically observable metric is blocking (rightmost three columns).

System	W_q (s) Analytical	Poisson DES	Empirical DES	Best-fit DES	P_{block} (%) Erlang B	Emp. DES	Observed
BB Kendall/MIT	0.8	1.4 ± 0.5	9.4 ± 1.7	10.6 ± 3.6	0.068	0.54	3.73
BB MIT Vassar	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0	2.88
MBTA Northbound	20.1	20.5 ± 0.5	4.2 ± 0.1	4.3 ± 0.2	—	—	—
MBTA Southbound	10.7	10.9 ± 0.2	0.9 ± 0.1	1.0 ± 0.1	—	—	—

C Supplementary figures

Figures 2–9 provide the full set of diagnostics summarised in the main text. The block structure is: phase-1 exploratory (2–4), phase-2 distribution fits (5–7), phase-3 queueing comparisons (8–9).

D Derivations

D.1 M/M/c waiting time

Let λ and μ denote the arrival rate and per-server service rate, $a = \lambda/\mu$, and c the number of servers, with the stability requirement $\rho = a/c < 1$. The Erlang C probability that an arrival finds all servers busy (and must queue) is

$$C(c, a) = \frac{\frac{a^c}{c!} \cdot \frac{1}{1-\rho}}{\sum_{k=0}^{c-1} \frac{a^k}{k!} + \frac{a^c}{c!} \cdot \frac{1}{1-\rho}}. \tag{3}$$

Conditional on queueing, the residual wait in an M/M/c system is exponential with rate $c\mu - \lambda$, so the unconditional expected wait is

$$W_q = \frac{C(c, a)}{c\mu - \lambda}. \tag{4}$$

The expected queue length follows by Little's Law: $L_q = \lambda W_q$; and the expected system time is $W = W_q + 1/\mu$.

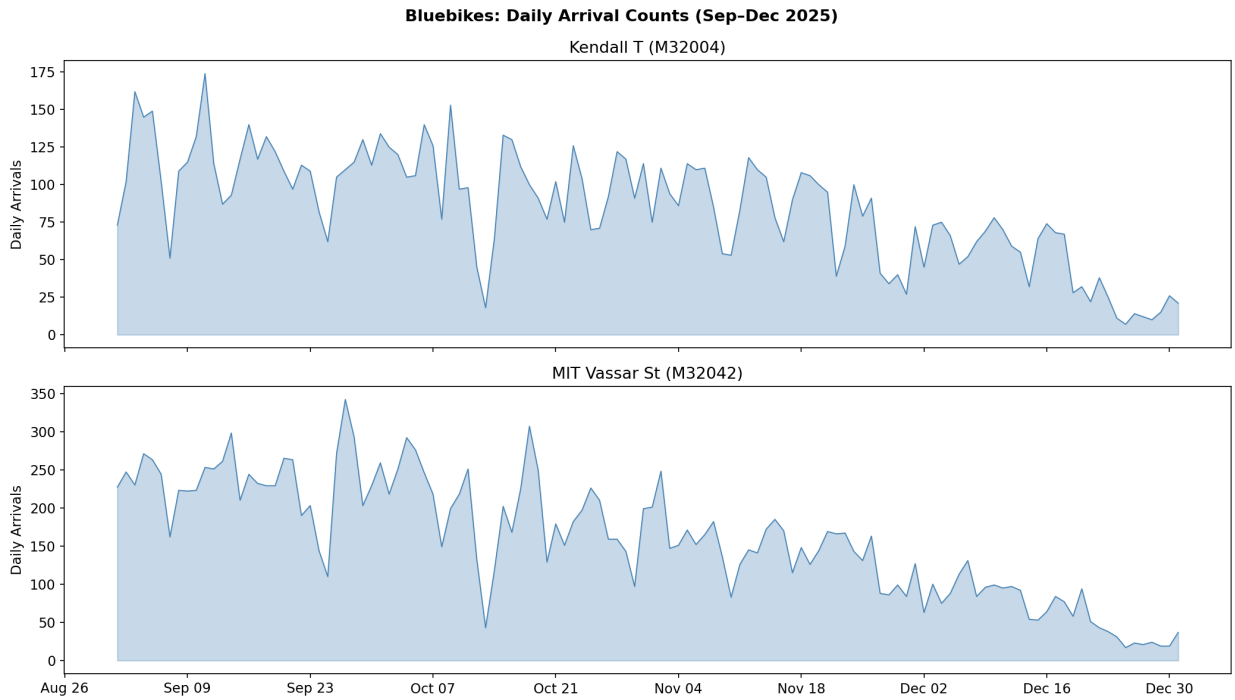

Figure 2: Bluebikes daily arrival counts at both stations, September–December 2025.

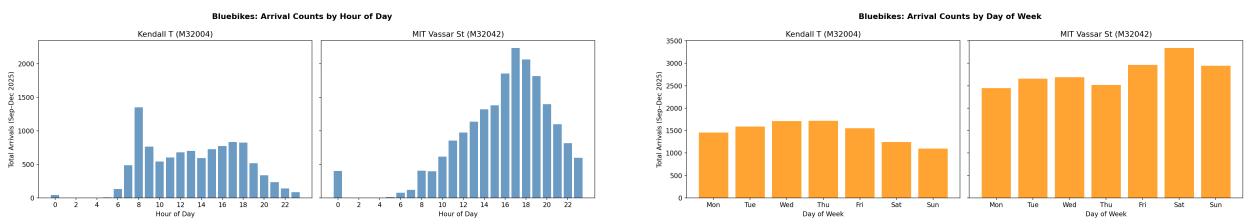

Figure 3: Bluebikes arrivals by hour-of-day (left) and by day-of-week (right).

D.2 Erlang B (M/M/c/c) blocking probability

The finite-capacity M/M/c/c model rejects any arrival that finds all servers busy. The stationary probability of rejection is the Erlang B blocking formula

$$B(c, a) = \frac{a^c/c!}{\sum_{k=0}^{c} a^k/k!}, \tag{5}$$

which satisfies the numerically stable recursion

$$B(c, a) = \frac{a\, B(c-1, a)}{c + a\, B(c-1, a)}, \qquad B(0, a) = 1. \tag{6}$$

We use the recursion throughout.

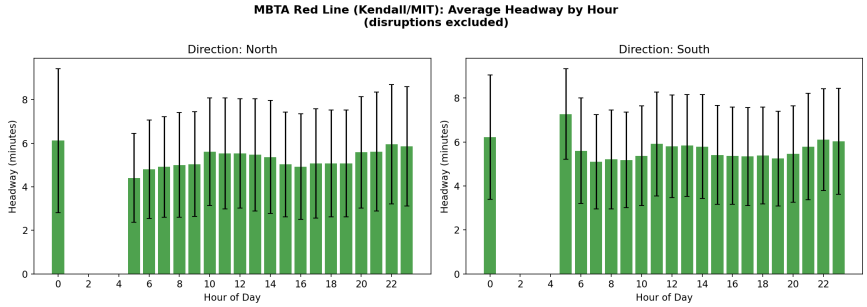

Figure 4: MBTA Red Line headway distribution by hour-of-day at Kendall/MIT.

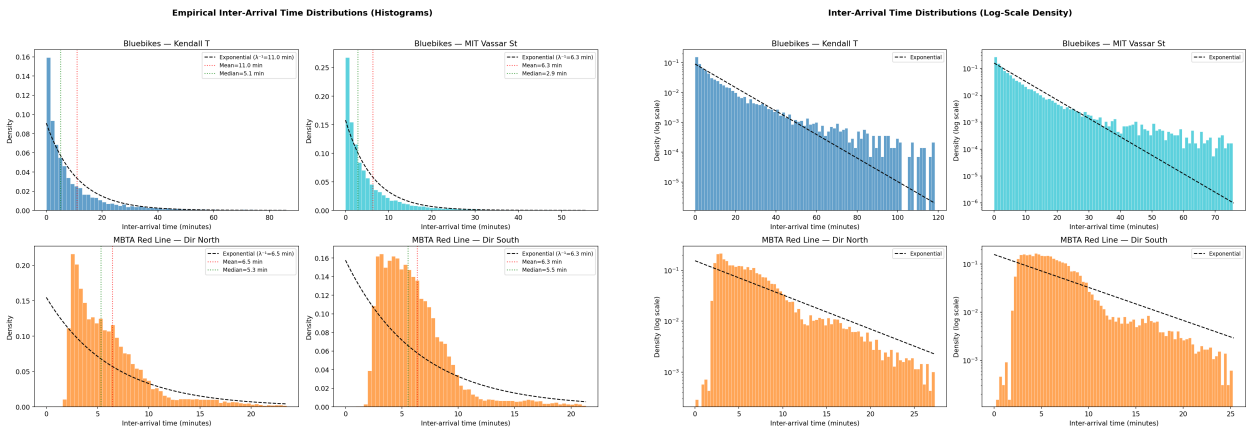

Figure 5: IAT histograms with fitted distributions (linear and log scale).

D.3 Bluebikes service rate via Little's Law

Little's Law [4] relates the mean number of entities in a stationary queueing system to the arrival rate and mean time spent: $L = \lambda W$. Applied to a single dock slot at station s during non-full intervals, $L_{\text{slot}} = L_{\text{non-full}}/c$ is the mean occupancy fraction of a slot, $\lambda_{\text{slot}} = \lambda/c$ the per-slot arrival rate, and $W_{\text{slot}} = 1/\mu$ the mean service time. Rearranging,

$$\mu = \frac{\lambda/c}{L_{\text{non-full}}/c} = \frac{\lambda}{L_{\text{non-full}}}. \tag{7}$$

This is a first-order estimate that treats all slots as exchangeable and conditions on non-full intervals to sidestep censoring. A more refined estimator would allow μ to vary by hour-of-day and is listed as future work (§7).

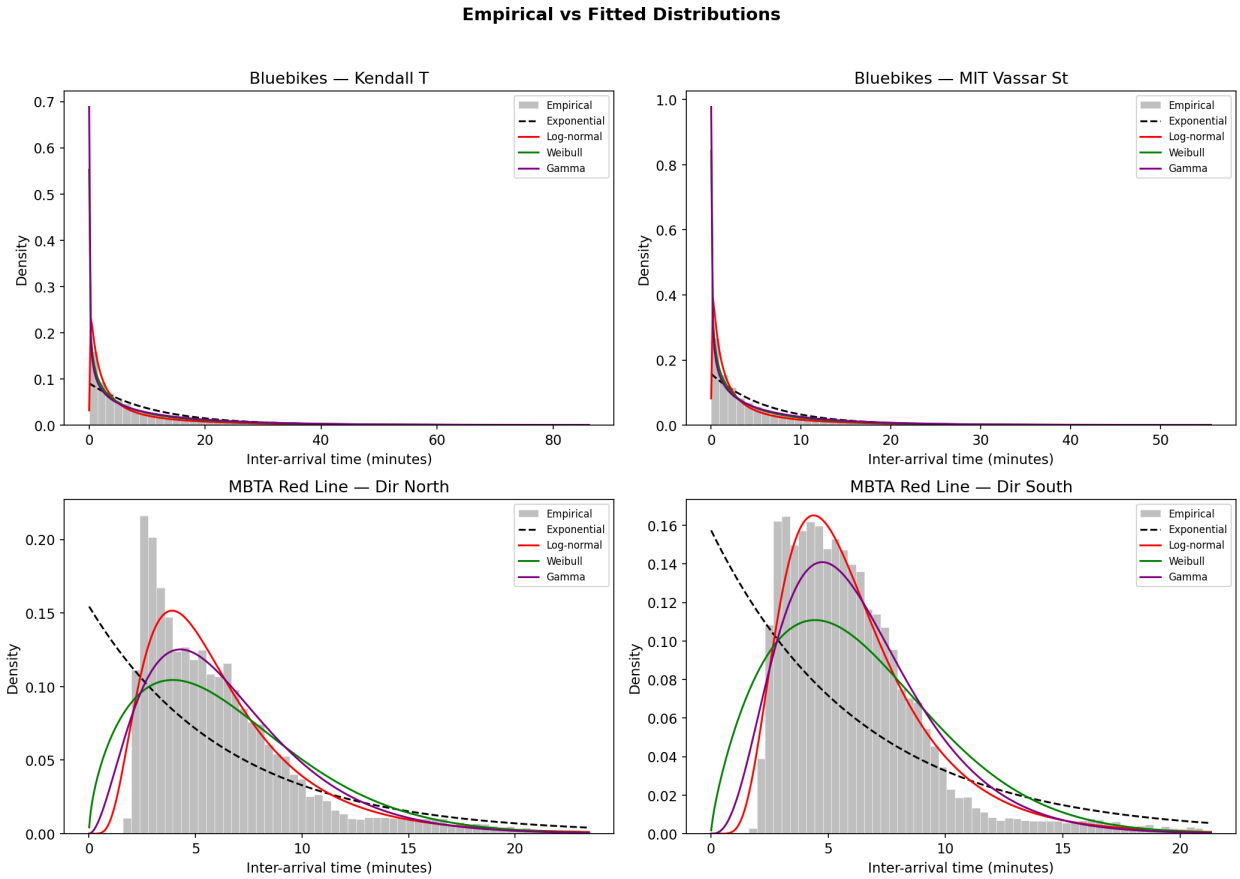

Figure 6: Fitted distributions overlaid on empirical PDFs (Weibull best for Bluebikes; log-normal best for MBTA).

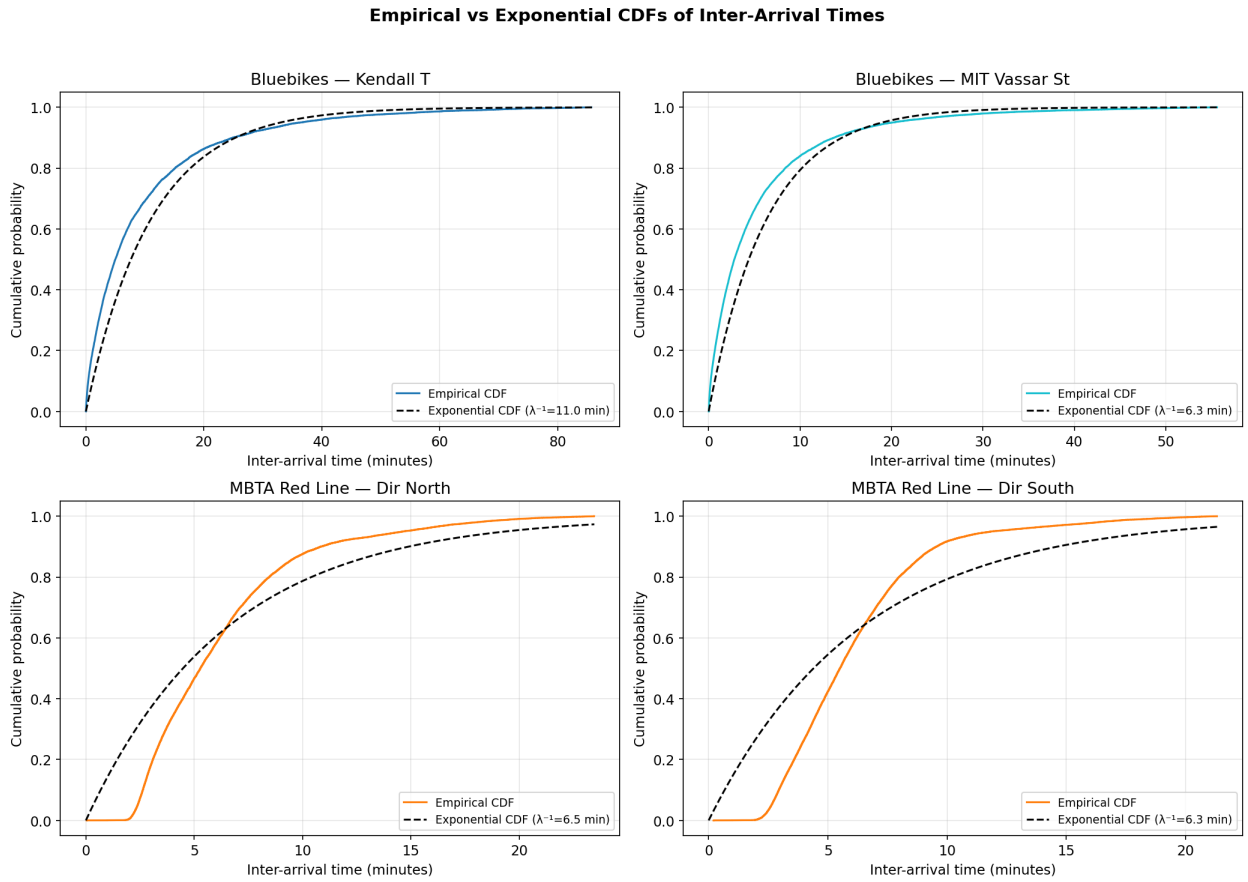

Figure 7: Empirical IAT CDFs with fitted exponential, Weibull, log-normal, and gamma overlays, for all four systems. Exponential (Poisson baseline) is visibly the worst fit in every panel.

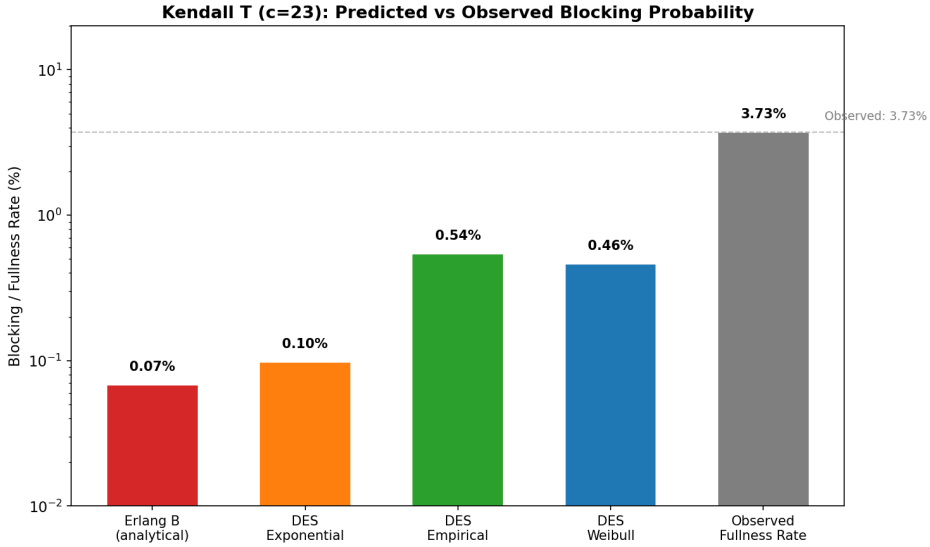

Figure 8: Blocking probability: Erlang B analytical vs. DES (Poisson) vs. DES (empirical) vs. observed full-capacity rate. Log y-axis. The 55× gap at Kendall/MIT is the visual centre; MIT Vassar St shows a qualitatively parallel gap from effectively zero predicted blocking to 2.88% observed.

Poisson Assumption Error Summary
*** Observed fullness rate (not from DES)**

System	Model	Metric	Poisson Prediction	Empirical DES	Relative Error	Direction
BB Kendall T	M/M/c	Wq (sec)	0.8	9.4	−91%	Underestimates
BB Kendall T	M/M/c	W (min)	122.8	123.0	−0.2%	≈ Correct
BB Kendall T	M/M/c/c	Block %	0.07%	0.54%	−87%	Underestimates
BB Kendall T	M/M/c/c	vs Observed	0.07%	3.73%*	−98%	Underestimates
BB MIT Vassar	M/M/c/c	vs Observed	≈0%	2.88%*	−100%	Underestimates
MBTA North	M/M/1	Wq (sec)	20.1	4.2	+376%	Overestimates
MBTA North	M/M/1	W (sec)	98.8	83.0	+19%	Overestimates
MBTA South	M/M/1	Wq (sec)	10.7	0.9	+1,050%	Overestimates
MBTA South	M/M/1	W (sec)	69.5	59.8	+16%	Overestimates

Figure 9: Relative error of analytical Poisson predictions against empirical-DES benchmarks, across systems and metrics.